KAIZEN MATHS

Lesson starter

Simultaneous Equations

Elimination Method

Name _____

S1

Solve the system:

$5x + 5y = 30$

$5x + 3y = 22$

S2

Solve the system:

$2x + 4y = 10$

$5x + 3y = 11$

S3

Solve the system:

$3x + 3y = 15$

$2x - 4y = -2$

S4

Solve the system:

$3x + 4y = 28$

$3x - 4y = -4$

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Simultaneous Equations

Elimination Method

Name _____

E1

Solve the system:

$2x + 3y = 20$

$5x + 5y = 40$

E2

Solve the system:

$5x + 5y = 25$

$3x - 5y = -17$

E3

Solve the system:

$6x + 3y = 27$

$6x - 2y = 22$

KAIZEN MATHS

Exit ticket

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$2x + 3y = 20$

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$6x + 3y = 27$

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Show your working

KAIZEN MATHS

Exit ticket

Simultaneous Equations

Elimination Method

Name _____

E1

Solve the system:

$2x + 3y = 20$

$5x + 5y = 40$

E2

Solve the system:

$5x + 5y = 25$

$3x - 5y = -17$

E3

Solve the system:

$6x + 3y = 27$

$6x - 2y = 22$

ALG-BE09-L2 · E

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Simultaneous Equations

Elimination Method

Name _____

E1

Solve the system:

$2x + 3y = 20$

$5x + 5y = 40$

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E3

Solve the system:

$6x + 3y = 27$

$6x - 2y = 22$

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Show your working

Teacher answers & working

Simultaneous Equations · Elimination Method

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S1

Solve the system:

$5x + 5y = 30$

$5x + 3y = 22$

Answer:

$x = 2, \; y = 4$

Working:

Step 1: Write the system.

$$\begin{cases} 5x + 5y = 30 \\ 5x + 3y = 22 \end{cases}$$

Step 2: Subtract equation (2) from equation (1) to eliminate x.

$5x + 5y - (5x + 3y) = 30 - 22$

$2y = 8$

Step 3: Solve for y, then back-substitute.

$y = 4 = 4$

$5x + 5 \times 4 = 30$

$x = 2 = 2$

Step 4: Check both equations.

$5 \times 2 + 5 \times 4 = 30$ ✓

$5 \times 2 + 3 \times 4 = 22$ ✓

Teacher answers & working

Simultaneous Equations · Elimination Method

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S2

Solve the system:

$2x + 4y = 10$

$5x + 3y = 11$

Answer:

$x = 1, \; y = 2$

Working:

Step 1: Write the system.

$$\begin{cases} 2x + 4y = 10 \\ 5x + 3y = 11 \end{cases}$$

Step 2: Make x-coefficients equal using the lowest common multiple.

$\text{LCM}\ (2, 5) = 10$

$5 \times (2x + 4y = 10) \Rightarrow 10x + 20y = 50$

$2 \times (5x + 3y = 11) \Rightarrow 10x + 6y = 22$

Step 3: Subtract the new equations to eliminate x.

$14y = 28$

$y = 2 = 2$

Step 4: Back-substitute for x.

$2x + 4 \times 2 = 10$

$x = 1 = 1$

Teacher answers & working

Simultaneous Equations · Elimination Method

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S3

Solve the system:

$3x + 3y = 15$

$2x - 4y = -2$

Answer:

$x = 3, \; y = 2$

Working:

Step 1: Write the system.

$$\begin{cases} 3x + 3y = 15 \\ 2x - 4y = -2 \end{cases}$$

Step 2: Make y-coefficients equal and opposite.

$\text{LCM}\ (|3|, |-4|) = 12$

$4 \times (3x + 3y = 15) \Rightarrow 12x + 12y = 60$

$3 \times (2x - 4y = -2) \Rightarrow 6x - 12y = -6$

Step 3: Add the new equations to eliminate y.

$18x = 54$

$x = 3 = 3$

Step 4: Back-substitute for y.

$3 \times 3 + 3y = 15$

$y = 2 = 2$

Teacher answers & working

Simultaneous Equations · Elimination Method

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S4

Solve the system:

$3x + 4y = 28$

$3x - 4y = -4$

Answer:

$x = 4, \; y = 4$

Working:

Step 1: Write the system.

$$\begin{cases} 3x + 4y = 28 \\ 3x - 4y = -4 \end{cases}$$

Step 2: Subtract equation (2) from equation (1) to eliminate x.

$3x + 4y - (3x + -4y) = 28 - -4$

$8y = 32$

Step 3: Solve for y, then back-substitute.

$y = 4 = 4$

$3x + 4 \times 4 = 28$

$x = 4 = 4$

Step 4: Check both equations.

$3 \times 4 + 4 \times 4 = 28$ ✓

$3 \times 4 + -4 \times 4 = -4$ ✓

Teacher answers & working

Simultaneous Equations · Elimination Method

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E1

Solve the system:

$2x + 3y = 20$

$5x + 5y = 40$

Answer:

$x = 4, \; y = 4$

Working:

Step 1: Write the system.

$$\begin{cases} 2x + 3y = 20 \\ 5x + 5y = 40 \end{cases}$$

Step 2: Make x-coefficients equal using the lowest common multiple.

$\text{LCM}\ (2, 5) = 10$

$5 \times (2x + 3y = 20) \Rightarrow 10x + 15y = 100$

$2 \times (5x + 5y = 40) \Rightarrow 10x + 10y = 80$

Step 3: Subtract the new equations to eliminate x.

$5y = 20$

$y = 4 = 4$

Step 4: Back-substitute for x.

$2x + 3 \times 4 = 20$

$x = 4 = 4$

Teacher answers & working

Simultaneous Equations · Elimination Method

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E2

Solve the system:

$5x + 5y = 25$

$3x - 5y = -17$

Answer:

$x = 1, \; y = 4$

Working:

Step 1: Write the system.

$$\begin{cases} 5x + 5y = 25 \\ 3x - 5y = -17 \end{cases}$$

Step 2: Make y-coefficients equal and opposite.

$\text{LCM}\ (|5|, |-5|) = 5$

$1 \times (5x + 5y = 25) \Rightarrow 5x + 5y = 25$

$1 \times (3x - 5y = -17) \Rightarrow 3x - 5y = -17$

Step 3: Add the new equations to eliminate y.

$8x = 8$

$x = 1 = 1$

Step 4: Back-substitute for y.

$5 \times 1 + 5y = 25$

$y = 4 = 4$

Teacher answers & working

Simultaneous Equations · Elimination Method

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E3

Solve the system:

$6x + 3y = 27$

$6x - 2y = 22$

Answer:

$x = 4, \; y = 1$

Working:

Step 1: Write the system.

$$\begin{cases} 6x + 3y = 27 \\ 6x - 2y = 22 \end{cases}$$

Step 2: Subtract equation (2) from equation (1) to eliminate x.

$6x + 3y - (6x + -2y) = 27 - 22$

$5y = 5$

Step 3: Solve for y, then back-substitute.

$y = 1 = 1$

$6x + 3 \times 1 = 27$

$x = 4 = 4$

Step 4: Check both equations.

$6 \times 4 + 3 \times 1 = 27$ ✓

$6 \times 4 + -2 \times 1 = 22$ ✓